

6. Write a program to implement k -Nearest Neighbour algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem.

```
import pandas as pd

from pandas import DataFrame

df_irisbd = pd.read_csv(r"/content/iris.data",header=None,index_col=None)
print(df_irisbd)

X = df_irisbd.iloc[:, :-1].values
y = df_irisbd.iloc[:, 4].values

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X,y,train_size=0.8,random_state=100)
print(y_test)

from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier(n_neighbors=3)

# Train the model using the training sets
model.fit(X_train,y_train)

predicted= model.predict(X_test) # 0:Overcast, 2:Mild

from sklearn.metrics import classification_report, confusion_matrix
print(confusion_matrix(y_test, predicted))
print(classification_report(y_test, predicted))
```